6. Data Visualization III

Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as:

1. **List down the features and their types (e.g., numeric, nominal) available in the dataset.**
2. **Create a histogram for each feature in the dataset to illustrate the feature distributions.**
3. **Create a box plot for each feature in the dataset. Compare distributions and identify outliers.**

In [2]:

```
import numpy as np
import seaborn as sns
import pandas as pd

sns.set_theme( rc={ "figure.figsize": (4,3) } )
```

In [3]:

```
ds = pd.read_csv( "iris.csv" )
ds.head()
```

Out[3]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

1. Feature Information

Numeric features: `sepal_width`, `sepal_length`, `petal_length` **and** `petal_width`

Nominal features: `species`

In [4]:

```
ds.dtypes
```

Out[4]:

```
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species          object
dtype: object
```

In [5]:

```
ds.describe()
```

Out[5]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000

	sepal_length	sepal_width	petal_length	petal_width
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

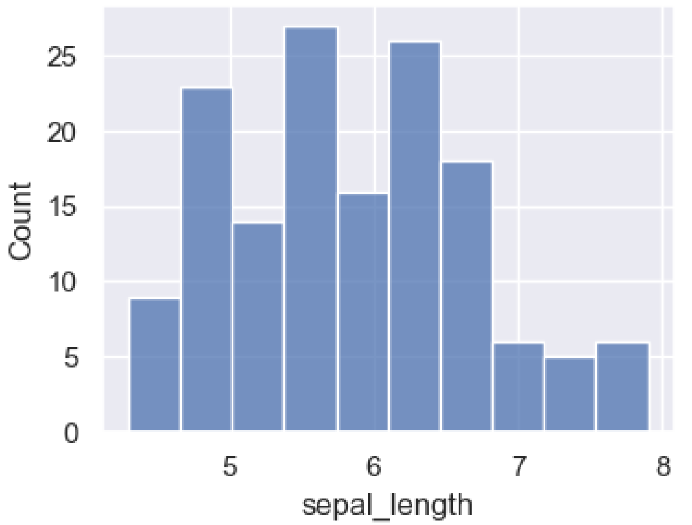
2. Feature Distribution with Histograms

In [6]:

```
sns.histplot( data=ds , x="sepal_length" , bins=10 )
```

Out[6]:

<Axes: xlabel='sepal_length', ylabel='Count'>

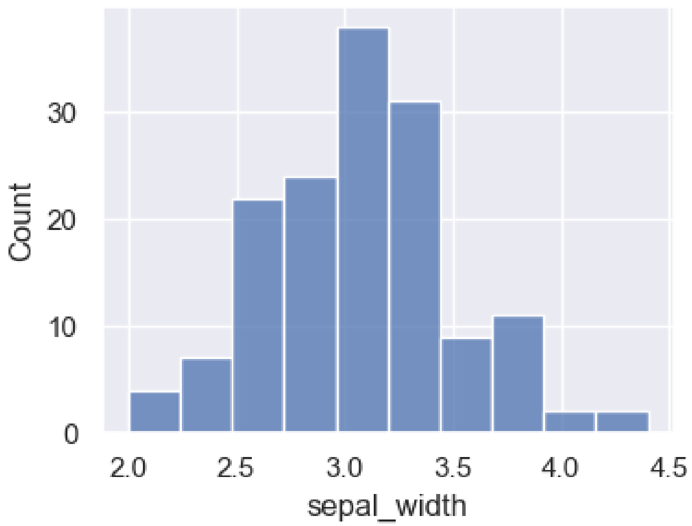


In [7]:

```
sns.histplot( data=ds , x="sepal_width" , bins=10 )
```

Out[7]:

<Axes: xlabel='sepal_width', ylabel='Count'>

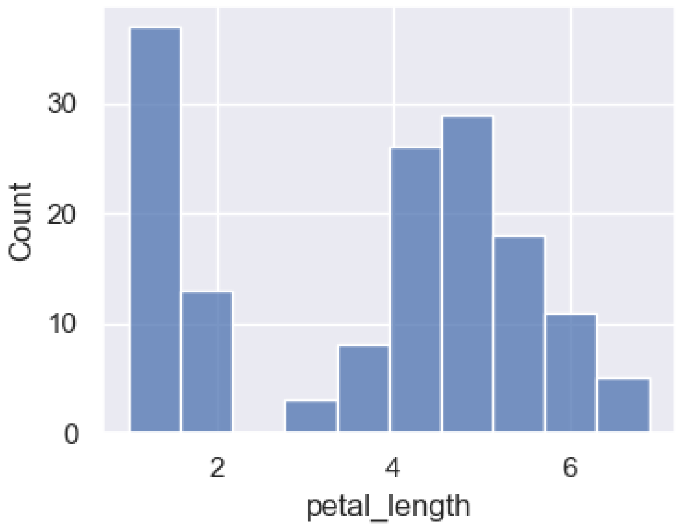


In [8]:

```
sns.histplot( data=ds , x="petal_length" , bins=10 )
```

Out[8]:

```
<Axes: xlabel='petal_length', ylabel='Count'>
```

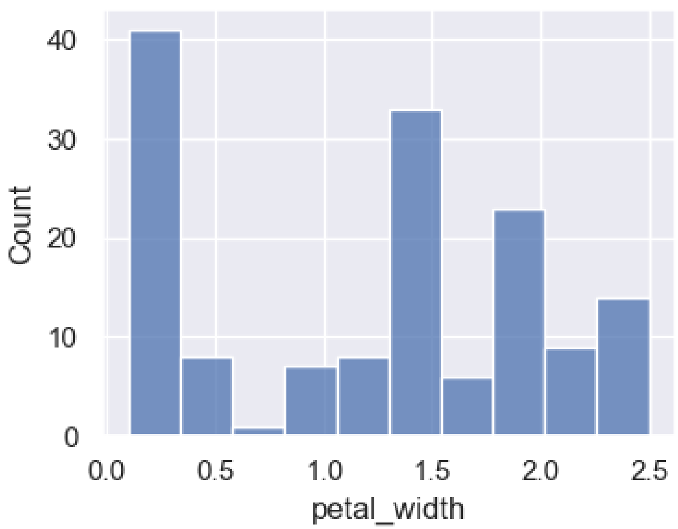


In [9]:

```
sns.histplot( data=ds , x="petal_width" , bins=10 )
```

Out[9]:

```
<Axes: xlabel='petal_width', ylabel='Count'>
```



3. Visualizing feature distribution with box plots

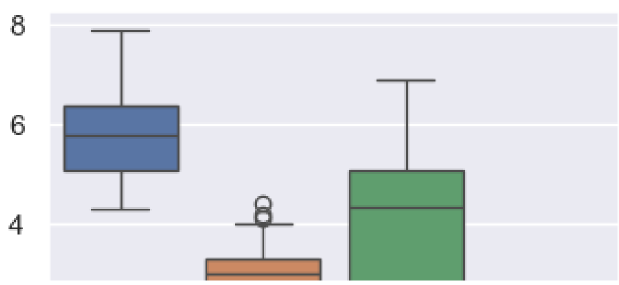
3.1. Visualizing the distribution of numeric features

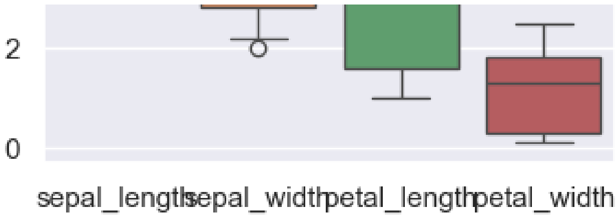
In [15]:

```
sns.boxplot( data=ds.drop( [ "species" ] , axis=1 ) )
```

Out[15]:

```
<Axes: >
```





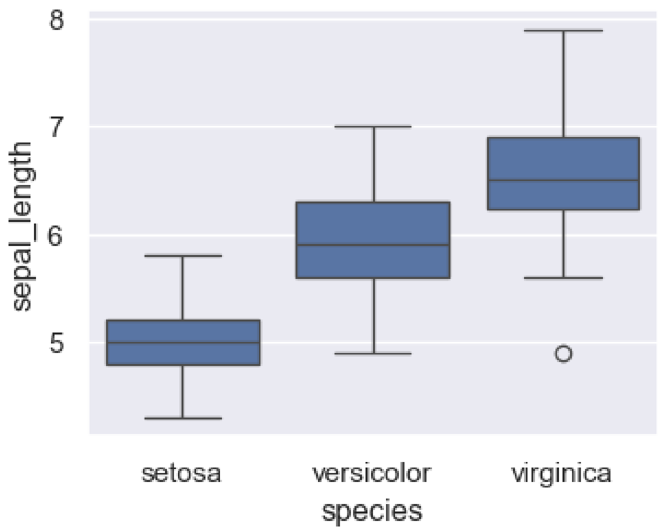
3.2. Visualizing the distribution of each numeric feature against `spec

In [16]:

```
sns.boxplot( data=ds , x="species" , y="sepal_length" )
```

Out[16]:

<Axes: xlabel='species', ylabel='sepal_length'>

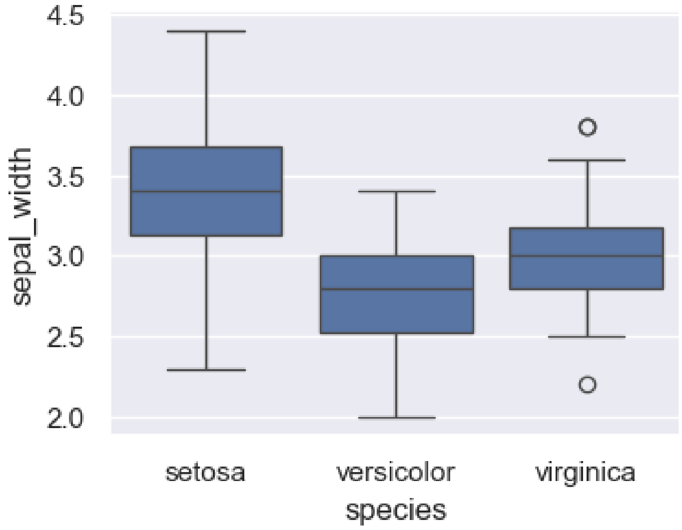


In [17]:

```
sns.boxplot( data=ds , x="species" , y="sepal_width" )
```

Out[17]:

<Axes: xlabel='species', ylabel='sepal_width'>

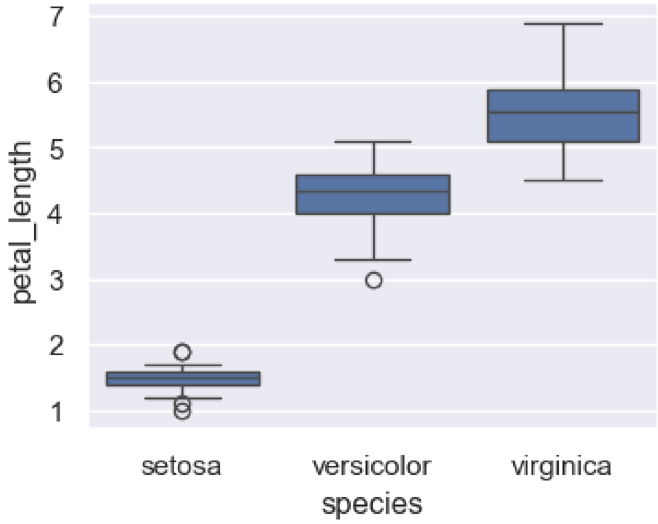


In [18]:

```
sns.boxplot( data=ds , x="species" , y="petal_length" )
```

Out[18]:

<Axes: xlabel='species', ylabel='petal_length'>

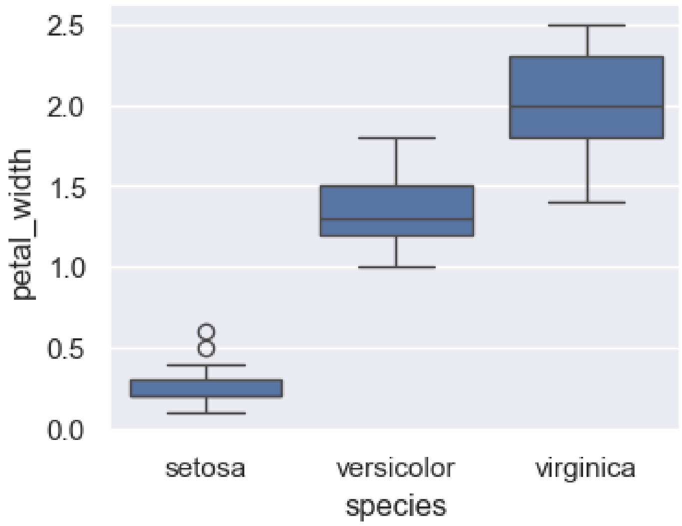


In [19]:

```
sns.boxplot( data=ds , x="species" , y="petal_width" )
```

Out[19]:

```
<Axes: xlabel='species', ylabel='petal_width'>
```



3.3. Comparing distributions

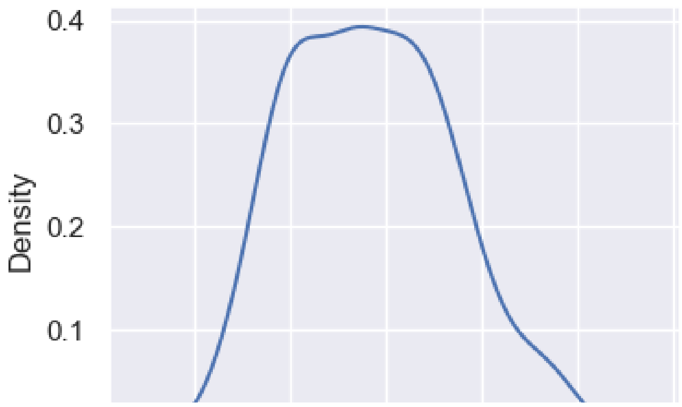
3.3.1. Comparing distributions visually with KDE plots

In [20]:

```
sns.kdeplot( ds , x="sepal_length" )
```

Out[20]:

```
<Axes: xlabel='sepal_length', ylabel='Density'>
```

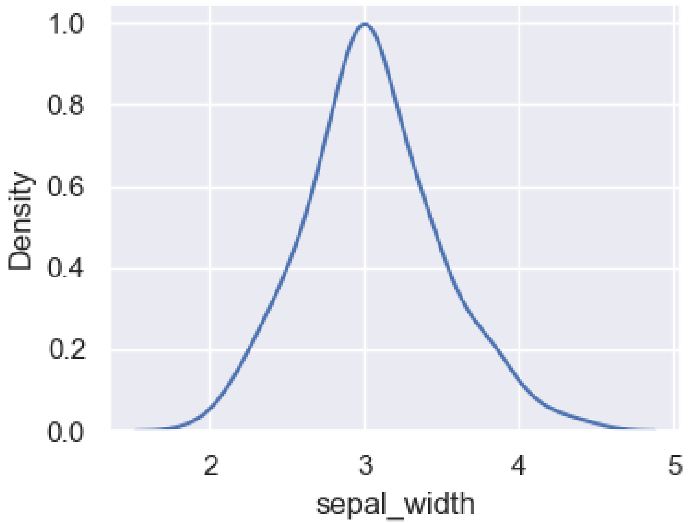


In [21]:

```
sns.kdeplot( ds , x="sepal_width" )
```

Out[21]:

```
<Axes: xlabel='sepal_width', ylabel='Density'>
```

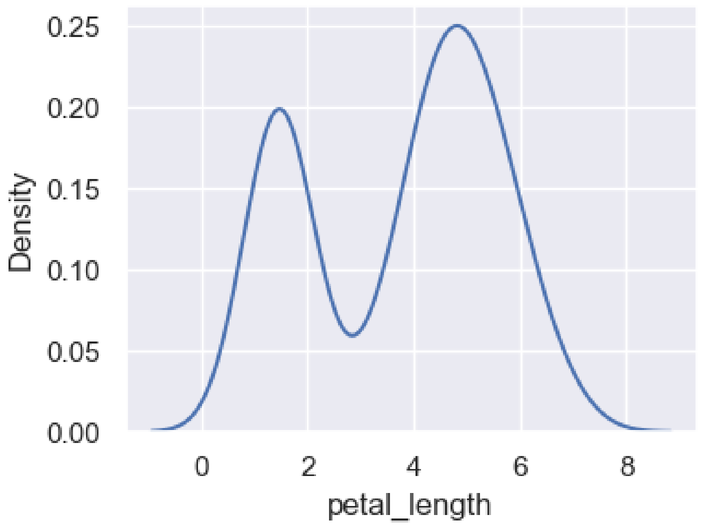


In [22]:

```
sns.kdeplot( ds , x="petal_length" )
```

Out[22]:

```
<Axes: xlabel='petal_length', ylabel='Density'>
```

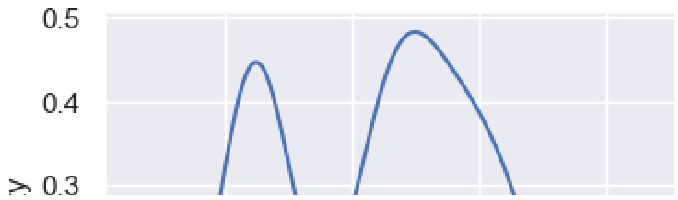


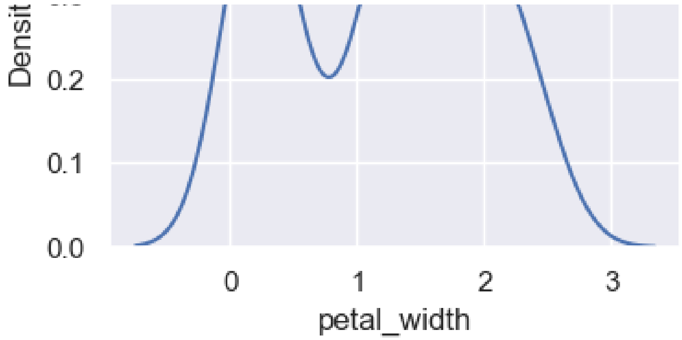
In [23]:

```
sns.kdeplot( ds , x="petal_width" )
```

Out[23]:

```
<Axes: xlabel='petal_width', ylabel='Density'>
```





3.3.2. Comparing distributions with Pearson's correlation

In [24]:

```
ds[ [ "sepal_length" , "sepal_width" , "petal_length" , "petal_width" ] ].corr()
```

Out[24]:

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	-0.109369	0.871754	0.817954
sepal_width	-0.109369	1.000000	-0.420516	-0.356544
petal_length	0.871754	-0.420516	1.000000	0.962757
petal_width	0.817954	-0.356544	0.962757	1.000000